in such an elaborate pattern, it embodies a huge amount of energy. Thus, all but the simplest string vibrational patterns are highly energetic and hence, via $E = mc^2$, correspond to particles with huge masses.

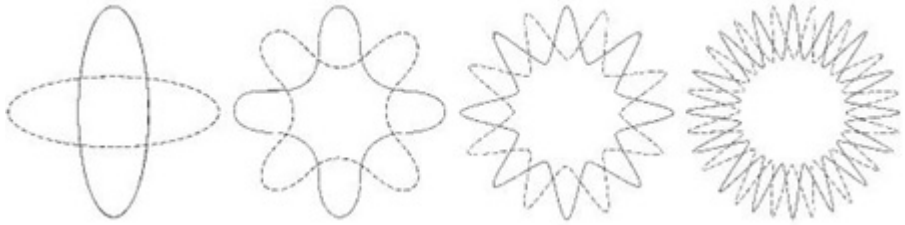


Figure 12.4 The first few examples of string vibrational patterns.

And by huge, I really mean huge. Calculations show that the masses of the string vibrations follow a series analogous to musical harmonics: they are all multiples of a fundamental mass, the *Planck mass,* much as overtones are all multiples of a fundamental frequency or tone. By the standards of particle physics, the Planck mass is colossal—it is some 10 billion billion (10^{19}) times the mass of a proton, roughly the mass of a dust mote or a bacterium. Thus, the possible masses of string vibrations are 0 times the Planck mass, 1 times the Planck mass, 2 times the Planck mass, 3 times the Planck mass, and so on, showing that the masses of all but the 0-mass string vibrations are gargantuan.[15]

As you can see, some of the particles in Tables 12.1 and 12.2 are indeed massless, but most aren't. And the nonzero masses in the tables are farther from the Planck mass than the Sultan of Brunei is from needing a loan. Thus, we see clearly that the known particle masses do not fit the pattern advanced by string theory. Does this mean that string theory is ruled out? You might think so, but it doesn't. Having an endless list of vibrational patterns whose masses become ever more remote from those of known particles is a challenge the theory must overcome. Years of research have revealed promising strategies for doing so.

As a start, note that experiments with the known particle species have taught us that heavy particles tend to be unstable; typically, heavy particles disintegrate quickly into a shower of lower-mass particles, ultimately generating the lightest and most familiar species in Tables 12.1 and 12.2. (For instance, the top-quark disintegrates in about 10^{-24} seconds.) We expect this lesson to hold true for the "superheavy" string